

Praveena Satkunarajah, PhD

psatkunaraja@mun.ca • Citizenship: Singapore

Auditory neuroscientist specializing in auditory perception and representation, decoding EEG using machine learning and statistical techniques to understand sensory representations

Education

Jan 20 – May 26	Memorial University of Newfoundland, Canada <i>PhD, Neuroscience</i> <i>Understanding encoding of musical timbre on the auditory pathway</i> <i>Dr Benjamin Zendel</i>
Aug 16 – Jun 19	Nanyang Technological University, Singapore <i>MSc in Information Studies</i>
Aug 10 – Jun 15	Nanyang Technological University, Singapore <i>(#6 CS program in global rankings)</i> <i>Bachelor of Engineering in Computer Science</i> Honours Project: Diagnosing ADHD based on fMRIs using Neural Networks

Work Experience

Memorial University of Newfoundland, Canada <i>Postdoctoral Fellow • Advisor: Dr Benjamin Zendel</i>	May 26 – Present
Memorial University of Newfoundland, Canada <i>Lab Manager</i>	Aug 23 – Present
National Research Council, Canada <i>Research Assistant</i>	Aug 23 – Apr 24
A * STAR Institute of High Performance Computing, Singapore <i>Research Engineer</i>	Jun 15 – Nov 19

Refereed Publications

Journal Articles

- **Satkunarajah, P.**, Power, S. D., & Zendel, B. R. (2026). Single-trial EEG-based classification reveals instrument-specific timbre perception via traditional machine learning classifiers. *NeuroImage*, 336, 122010. doi:[10.1016/j.neuroimage.2026.122010](https://doi.org/10.1016/j.neuroimage.2026.122010)
- Sauv , S.A., **Satkunarajah, P.**, & Zendel, B.R. (2023). Tracking the emergence of a pitch hierarchy using an artificial grammar requires extended exposure. *Frontiers in Cognition*, 2(1027259), 1-14. doi: [10.3389/fcogn.2023.1027259](https://doi.org/10.3389/fcogn.2023.1027259)
- Sauv , S. A., **Satkunarajah, P.**, Cooke, S., Demirkaplan,  ., Follett, A., & Zendel, B. R. (2024). Age and familiarity effects on musical memory. *PloS one*, 19(7), e0305969. doi: [10.1371/journal.pone.0305969](https://doi.org/10.1371/journal.pone.0305969)

Conference Proceedings

- **Satkunarajah, P.**, Power, S.D and Zendel, B.R, Decoding Musical Timbre Perception from Single-Trial EEG Data, 2024 IEEE International Conference on Systems, Man, and Cybernetics (SMC), Kuching, Malaysia, 2024, pp. 2016-2020, doi: [10.1109/SMC54092.2024.10831757](https://doi.org/10.1109/SMC54092.2024.10831757)
- Herberg, J. S., Yengin, I., **Satkunarajah, P.**, & Tan, M. (2017). Boosting Knowledge-Building with Cognitive Dialog Games. In *CogSci*.

Publications In Prep

- **Satkunarajah, P.**, Zendel, B.R., Effect of timbral priming on low-level physiology explored using Frequency Following Responses, *JNeurosci*, *revisions invited*
- **Satkunarajah, P.**, Zendel, B.R., Effect of timbral priming on low-level physiology in Older Adults, *Neurobiology of Aging*, *Under peer review*

Preprints

- Satkunarajah, P., & Agres, K. (2021). Web Platform for Visualisation of Kinematic Data captured from a Motor Tele-rehabilitation System. arXiv preprint arXiv:[2112.05118](https://arxiv.org/abs/2112.05118)

Conference Presentations

- **Satkunarajah, P.**, Kumar, M. G (*co-first author*), Zendel, B.R. (2026), Reservoir Network Model of Asymmetric Acoustic Priming reveals Chaotic Dynamics in Auditory Brainstem Model, *Computational Cognitive Neuroscience 2026*, NY, US
- **Satkunarajah, P.**, Zendel, B.R. (2026), Timbral context modulates the Frequency Following Response. *Annual Meeting of the Cognitive Neuroscience Society, Vancouver, BC*
- **Satkunarajah, P.**, Zendel, B.R. (2025) Priming effects on Timbre Encoding in auditory brainstem responses. *Biomedicine Symposium 2025*, St. John's, NL, CA
- **Satkunarajah, P.**, Power, S.D & Zendel, B.R. (2024). Decoding musical timbre perception from single-trial EEG data. *IEEE Systems, Man and Cybernetics (SMC) Kuching, Malaysia*
- **Satkunarajah, P.**, Power, S.D., & Zendel, B.R. (2024). Exploring timbre coding in EEG using machine learning. *Annual meeting of the Cognitive Neuroscience Society, Toronto, ON*
- **Satkunarajah, P.** & Zendel, B. (2023). Decoding Timbre Perception using Machine Learning. Talk presented at Canadian Society for Brain, Behaviour and Cognitive Science (CSBBCS) *Guelph, CA*
- Sauv , S.A, **Satkunarajah, P.**, Cooke, S., Demirkaplan,  ., Follett, A., & Zendel, B.R. (2023) Age and familiarity effects on musical memory. *International conference on music perception and cognition, Tokyo, Japan & online*
- Sauv , S.A, **Satkunarajah, P.** & Zendel, B.R. (2023). Tracking the emergence of a pitch hierarchy using an artificial grammar requires extended exposure. *International conference on music perception and cognition, Tokyo, Japan & online*
- **Satkunarajah, P.** & Zendel, B.R. (2022) Classifying the perception of different instruments using single trial EEG. *Annual meeting of the Canadian Acoustical Association, St. John's, NL*
- **Satkunarajah, P.** & Zendel, B.R. (2022). Using machine learning to classify EEG data based on timbre. *Society of Music Perception and Cognition Conference, Portland, USA*
- **Satkunarajah, P.** & Agres, K. R. (2021). Web-platform Prototype for Visualisation of Kinematic Data captured from a Music-based Tele-rehabilitation System. *International Conference on Music Perception and Cognition (ICMPC), Sheffield, UK (Virtual)*

Ongoing Projects

Praveena Satkunarajah, Grace Fitzgerald, Benjamin Zendel. Effects of musicianship on context-dependent neural representations in the auditory brainstem

Praveena Satkunarajah, M Ganesh Kumar, Sarah Power, Benjamin Zendel. Modelling context dependence in subcortical auditory circuits

Danica Robichaund, Rezvan Karaji, Praveena Satkunarajah, Benjamin Zendel. Neural representations of perceived musical goodness-of-fit

Teaching

Teaching Assistantship

Statistics and AI for Healthcare (*Material Development*)

Practical Labs

UGME Neuroanatomy Labs (2025) • Undergraduate EEG Labs (2022-2024)

Student Supervision

Grace Fitzgerald – *Research Assistant, Co-supervisor, Canada*

Bhavya Sehgal – *Undergraduate Honours, Co-supervisor, Canada*

A.M. Aditya – *MTech Placement, Co-supervisor, Singapore*

Harikrishnan Rajkumar – *MTech Placement, Co-supervisor, Singapore*

Invited Talks

2026 **Oregon Hearing Research Centre, Oregon Health and Sciences University, US**

Neural Representations of Musical Timbre Across the Auditory Pathway,

2026 **Centre for Research on Brain, Language and Music, McGill University, CA**

Context Dependent Modulation in Auditory Brainstem Responses,

2025 **School of Communication Sciences and Disorders, UCF, US**

Single-trial EEG Classification of Timbre Perception,

2025 **Brainwaves, Memorial University of Newfoundland, NL, CA**

Context-dependent neural encoding in the auditory brainstem.

2025 **Niagara Falls Memorial Medical Centre, New York, US**

AI in healthcare,

2024 **Brainwaves, Memorial University of Newfoundland, NL, CA**

Decoding musical timbre perception using single-trial EEG,

Media Highlights

2024 **Scientific American, Musical Memories Don't Fade with Age**

Awards & Honours

Dissertation Completion Award - \$5000, IEEE SMC Travel Award, 2024 - \$500, Dean's Building a Healthy Tomorrow Travel Award, 2024 - \$800, Nomination for Best Paper Award, ICMPC, 2021, SGS Graduate Fellowship, 2020 – 2025 - \$30,000

Society Memberships

CNS (2024, 2026), SMPC (2022 – 2024), IEEE (2024), CSBBCS (2023)

Skills

Lab skills: EEG acquisition and analysis, Computational Modelling, Machine Learning, Statistics, Acoustic Stimulus Design, Signal Processing, Data Visualisation, MoCAP

Programming Languages: Java, C#, Python, MATLAB, Perl, R, Javascript, SQL

Software Toolkits: scikit-learn, numpy, scipy, pytorch, Tensorflow, Unity, REAPER

Service

Science Outreach and Communication

WISE Summer Program (2026), Sparks Group (2024), Soapbox Science (2022-2023), *Annual meeting of the Canadian Acoustical Association* (2022) – Organizing Committee

Peer Review

IEEE SMC (2024)

Besides Academics

Sound Symposium (2024, 2026), ONSOUND (2025) – St. John's, NL, CA, Mental Act SG, Research Team (2021 – 2023)

References

Dr. Benjamin Zendel: bzendel@mun.ca

Dr. Sarah Sauve: ssauve@lincoln.ac.uk

Dr. Sarah Power: b09sdp@mun.ca